CHITIPA EDUCATION CLUSTER

MARKING SCHEME- MATHEMATICS II

SECTION A (60 marks)

1. Factorise completely $3a^3 - 75a$

$=3a(a^2 - 25)$ $\qquad$ $=3a^3 - 15a^2 + 15^a - 75a$

$=3a(a^2 - 5^2)$ $\quad$ OR $\quad$ $=3a^2(a - 5) + 15a(a - 5)$

$=3a(a - 5)(a + 5)$ $\qquad$ $=3a(a - 5)(a + 5)$

$=3a(a - 5)(a + 5)$

(3 marks)

2. a. Given that $\sin\theta = \frac{1}{\sqrt{5}}$ .Calculate, without the use of calculator the value of $\tan\theta$

$x^2 + 1^2 = (\sqrt{5})^2$ pyth thm $\qquad$ Since $\frac{\sin\theta}{\cos\theta} = \tan\theta$

$x^2 + 1 = 5$ $\qquad$ $\tan\theta = \frac{1}{\sqrt{5}} \div \frac{2}{\sqrt{5}}$

$x^2 = 4$ $\qquad$ OR $\qquad$ $= \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{2}$

$x = 2$ $\qquad$ $\tan\theta = \frac{1}{2}$

$\tan\theta = \frac{1}{x}$

$= \frac{1}{2}$ or 0.5

(4 marks)

b. Use the remainder theorem to show that w-m is a factor of the polynomial $w^2(m-2) + m^2(2-w)$

let $w-m = 0$

$\therefore w = m$

$\therefore w^2(m-2) + m^2(2-w) \rightarrow m^2(m-2) + m^2(2-m)$

$\rightarrow m^3 - 2m^2 + 2m^2 - m^3$

$\rightarrow 0$

$\therefore$ w -m is a factor(remendar theorem)

OR

Substitute m for w

i.e. $= w^2(w-2) + w^2(2-w)$

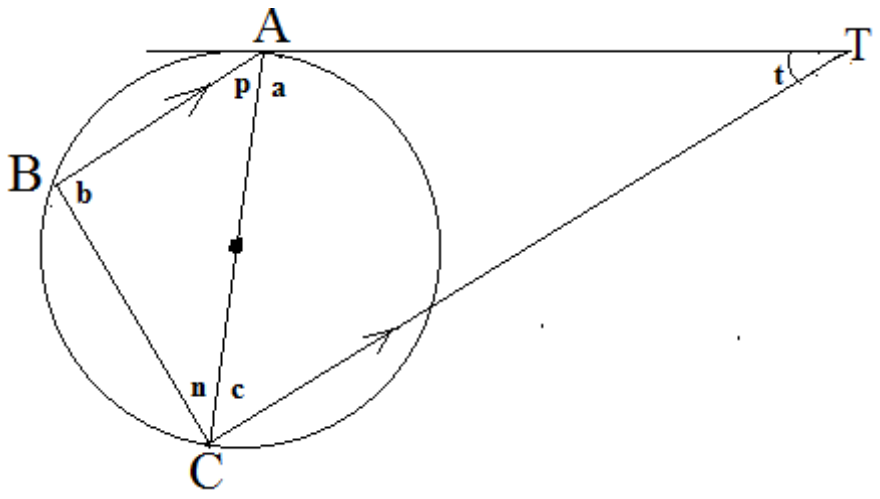
$= w^3 - 2w^2 + 2w^2 - w^3$

$= 0$

$\therefore$ w -m is a factor (remainder theorem)

(4 marks)

3. a. In the figure below in which AT is a tangent, CT is parallel to AB and AC is a diameter, show that triangle ABC is similar to triangle ATC



From Δs ABC and CAT

angle CAT =angle ABC(rt angles/)

angle BAC =angle ACT(alt.angles)

anle BAC =angle ATC(3rd angle)

∴ΔABC///ΔATC(AA)

OR

ref. notations angle b=angle a(angles in alt seg.)

angle p=angle c(alt. angles AB//TC)

angle n=t(3rd angle)

∴ΔABC///ΔATC(AA)

(4 marks)

b. Solve the equation $2^{2y+1} + 8 = 17(2^y)$

$(2^y)^2 \times 2 + 8 = 17(2^y)$

Let $2^y = x$

$$2x^2 - 17x + 8 = 0$$

$$2x^2 - 16x - x + 8 = 0$$

$$2x(x - 8) - (x - 8) = 0$$

$$(x - 8)(2x - 1) = 0$$

$$x = 8 \text{ or } \frac{1}{2}$$

$$2^y = 2^3 \text{ or } 2^{-1}$$

$$y = 3 \text{ or } -1$$

(6 marks)

4. a. If $f(x) = x^2 - 2x + 8$, find all the values of K given that $f(k + 1) = 8$

Since $f(k + 1) = 8$

$\therefore (k + 1)^2 - 2(k + 1) + 8 = 8$

$k^2 + 2k + 1 - 2k - 2 = 0$

$k^2 - 1 = 0$

$k^2 = 1$

$k = \pm 1$

OR

$k^2 - 1 = 0$

$(k - 1)(k + 1) = 0$

$k = 1 \text{ or } -1$

(5 marks)

b. A car starts from rest and accelerates uniformly to 4m/s in 6 seconds. It

further accelerates uniformly to a velocity of 90m/s in the next 3 seconds.

It maintains this velocity for 2 seconds. Using a scale of 2cm to 20m/s on

vertical axis and 2cm to 2 sec on horizontal axis, draw the speed-

time

graph for this car. **(5 marks)**

5. a. If A(h,3) and B(4,e) has its mid point M(3,5) on AB. Find the values of h

and e.

$m = \frac{1}{2}(x_2 + x_1, y_2 + y_1)$

$(3,5) = \frac{1}{2}(4 + h, 3 + e)$

$(6,10) = (4^{th}, e + 3)$

$3 = \frac{h + 4}{2}$ and $5 = \frac{e + 3}{2}$

$\therefore 6 = h + 4, \quad e + 3 = 10$

OR $h = 2, \quad e = 7$

$(6 - 4, 10 - 3) = (h,e)$

$(2,7) = (h,e)$

$\therefore h = 2$ and $e = 7$

(4 marks)

b. Find an equation of a parabola that has roots -3 and 2, and passes through the point $(-2, 6)$ in the form $y = ax^2 + bx + c$

$$\text{let } y = a(x+3)(x-2)$$

$$= a(x^2 + x - 6)$$

$$6 = a((2)^2 + (2) - 6$$

$$6 = a(4 - 2 - 6)$$

$$6 = -4a$$

$$a = -\frac{3}{2} \qquad y = -\frac{3}{2}(x^2 + x - 6)$$

$$= -\frac{3x^2}{2} - \frac{3x}{2} + 9$$

$$\text{or } = -1.5x - 1.5x + 9$$

(5 marks)

c. Show that $\begin{pmatrix} 5 & 0 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} -3 & 2 \\ 4 & -1 \end{pmatrix} - 5\begin{pmatrix} -3 & 2 \\ -2 & 1 \end{pmatrix}$ is a zero matrix

$$\text{LHS} = \begin{bmatrix} -15 + 0 & 10 + 0 \\ -6 - 4 & 4 + 1 \end{bmatrix} - \begin{bmatrix} -15 & 10 \\ -10 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -15 & 10 \\ -10 & 5 \end{bmatrix} - \begin{bmatrix} -15 & 10 \\ -10 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 5 & 0 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} -3 & 2 \\ 4 & -1 \end{bmatrix} - 5\begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix}$$

is a zero matrix since (LHS =0)

(4 marks)

6. a. Given that $2\log(x^2 - 5) = 4\log 2$. Calculate all values of x

$$\log (x^2 - 5)^2 = \log 2^4$$

$$(x^2 - 5)^2 = 2^4$$

$$x^2 - 5 = \pm 4$$

$$x^2 = \pm 4 + 5 = 9 \text{ or } 1$$

$$x = \pm 3 \text{ or } \pm 1$$

(5 marks)

b. The sum of the first n terms of an Arithmetic progression is $n(n + 2)$.

Calculate its 11th term

$$\text{let } n = 1 \rightarrow 1(1 + 2)$$

$$\therefore n = 3$$

$$\text{let } n = 2 \rightarrow 2(2 + 2)$$

$$\text{sum of 2 term} = 8$$

$$a + d = 8$$

$$3 + d = 8$$

$$d = 5$$

$$\therefore 11\text{th term} = a + 9d \text{ since } l = a + (n - 1)d$$

$$= 3 + 9(5)$$

$$= 48$$

(5 marks)

c. Solve for m and k in the simultaneous equation if $m^2 - k^2 = 39$

$$m - k = 3$$

$$\text{let } m = k + 3$$

$\therefore (k+3)^2 - k^2 = 39$

$k^2 + 6 + 9 - k^2 = 39$

$6k = 30$

but $k = \pm 5$

$\therefore m = 5 + 3$

$= \pm 8$ **(6 marks)**

SECTION B (40 marks)

Answer any 4 questions only

7. a. Copy and complete the table below if $y = x^3 - 2x^2 - 6x$

x	-2	-1	0	1	2	3	4
y	-4	**-3**	0	-7	-12	-9	**8**

b. Using a scale of 2 cm to 1 unit on the x-axis and 2cm to 4 units on the y-

axis, draw the graph for $y = x^3 - 2x^2 - 6x$

c. Use your graph to solve $x^3 - 2x^2 - 7x + 2 = 0$ **(10 marks)**

8. The resistance to motion of a car is partly constant and partly varies as the square of its speed. At 40km/hr, the resistance is 9860N and at 60km/hr, the resistance is 21860N.Calculate the speed at which resistance is 15260N \

let resistance be r and let speed be k

$r \alpha + \alpha k^2$

$r = a + bk^2$ **(where a and b are constant)**

$9860 = a + 40^2 b$

9860 =a +1600b.....(i)

also 21860 =a +3600b.....(ii)

$\therefore$12000 =2000b

b =6

a =9860 -9600

=260

r =260 +6k^2

15260 =260 +6k^2

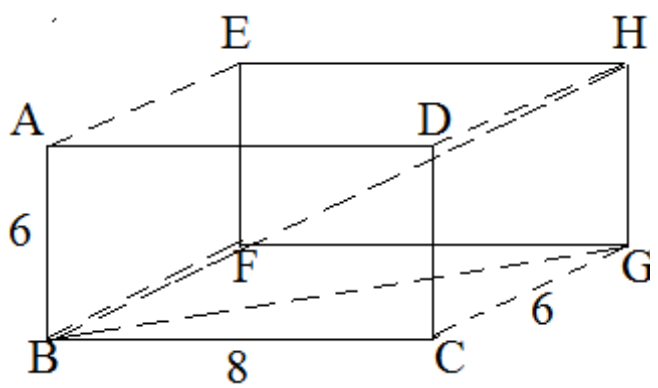
15000 =6k^2

2500 =k^2

k = ±50

k =50km/hr (10 marks)

9. In the cuboids below, all faces are rectangular. All dimensions are in cm



Calculate (i) length CE leaving your answer in simplified surd form

CE =BH(longest diagonals)

$$BH = \sqrt{BC^2 + CG^2 + GH^2} \qquad \text{(pyth.thm twice)}$$

$$= \sqrt{8^2 + 6^2 + 6^2}$$

$$= \sqrt{136}$$

$$= \sqrt{4 \times 34}$$

$$CE = 2\sqrt{34}\text{ cm}$$

(ii) angle between BH and the plane BCGF, correct to 1 decimal

place

$$BG^2 = 8^2 + 6^2 \text{(pyth.thm)}$$

$$= 100$$

$$BG = 10$$

When BG and BH are joined

$$\tan\theta = \frac{GH}{BG}$$

$$= \frac{6}{10}$$

$$\tan^{-1}\theta = 30.963$$

$\therefore$ angle between BH and plane BCGF $= 31.0^0$ correct to 1 decimal places

(10 marks)

10. Given that $t^2 - 3t + 2$ is a factor of $t^3 + at^2 + kt - 6$, calculate the values of a and k hence find the other factor

$$t^2 - 3t + 2 = (t-2)(t-1) = 0$$

$$\therefore t = 2 \text{ or } 1$$

$$\text{for } t \to 2 \text{ gives } 2^3 + 2^2 a + 2k - 6 = 0$$

$$4a + 2k = -2$$

$$2a + k = -1 \ldots\ldots (i)$$

$$\text{for } t \to 1 \text{ gives } 1 + a + + k - 6 = 0$$

$$a + k = 5 \ldots\ldots (ii)$$

$$(i) - (ii) \text{ has } a = -6, k = 11$$

$$(t^2 - 3t + 2)(at + x) \equiv t^3 - 6t^2 + 11t - 6$$

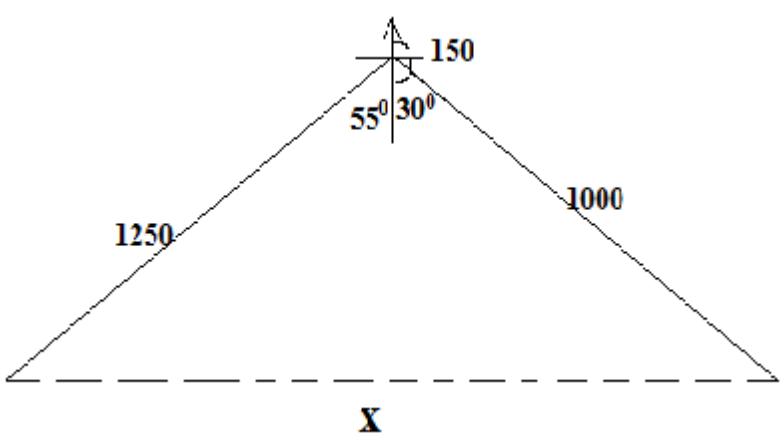
$$2x = -6$$

$$x = -3 \text{ and } at^3 = t^3$$

$$a = 1$$

$\therefore$ other factor $= (t - 3)$ **(10 marks)**

11. a. Two aeroplanes leave an airport at the same time. One flies on a bearing of 150^0 at 400km/hr and the other flies on the bearing of 235^0 at 500km/hr. How far apart are the two aeroplanes after $2\frac{1}{2}$ hours



$$2.5 \times 400 = 1000\text{km}$$

$$2.5 \times 500 = 1250\text{km}$$

$$x^2 = 1000^2 + 1250^2 - 2(1000)(1250)\cos 85$$

= (7 marks)

b. A certain translation maps P(3,5) into P'(7,8).Find the coordinates of Q if it is mapped into Q'(2,7) under same translation

P +Trans 1 = P'

$$\begin{pmatrix}3\\5\end{pmatrix} + \begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}7\\8\end{pmatrix}$$

$$\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}4\\3\end{pmatrix}$$

$$\therefore Q + \begin{pmatrix}4\\3\end{pmatrix} = \begin{pmatrix}2\\7\end{pmatrix}$$

Coord of 4 =(-2,4) **(3 marks)**

12. A trader would like to buy new cars for family use. There are two types of cars available: China and Japan. China cost K60000 each while Japan cost K100000 each. He has space to store only 15 cars and has only K1200000 available to spend.He expects to make a profit of K10000 for China and K15000 for Japan cars.He would like to spend atleast K400000 on Japan cars and atmost K600000 on China.

i. write down 4 inequalities that satisfy above information in addition to $x \geq 0$ and $y \geq 0$

given that x is number of china cars
and y is number of japan cars
$\therefore 60000x + 100000y \leq 1200000$
$3x + 5y \leq 60$....i(0,12);(20,0)
$60000x \leq x + y \leq 15$.....ii
$100000y \geq 400000$ (0,15);(15,0)
$y \geq 4$....iii

$x \leq 10$...iv

ii. using a scale of 2cm to 4,graph the above inequalities by shading unwanted region

iii. how many cars of each type can this trader buy to have greatest profits

P =10000x +15000y

A =0 +15000(4) =60000

B =10(10000) +4(15000) =160000

C =10(10000) +6 ×15000 =190000

D =5(10000) +1015000 =200000

E =0(10000) +10(15000) =150000

∴buy 5 china and 10 japan (10 marks)

END OF QUESTION PAPER